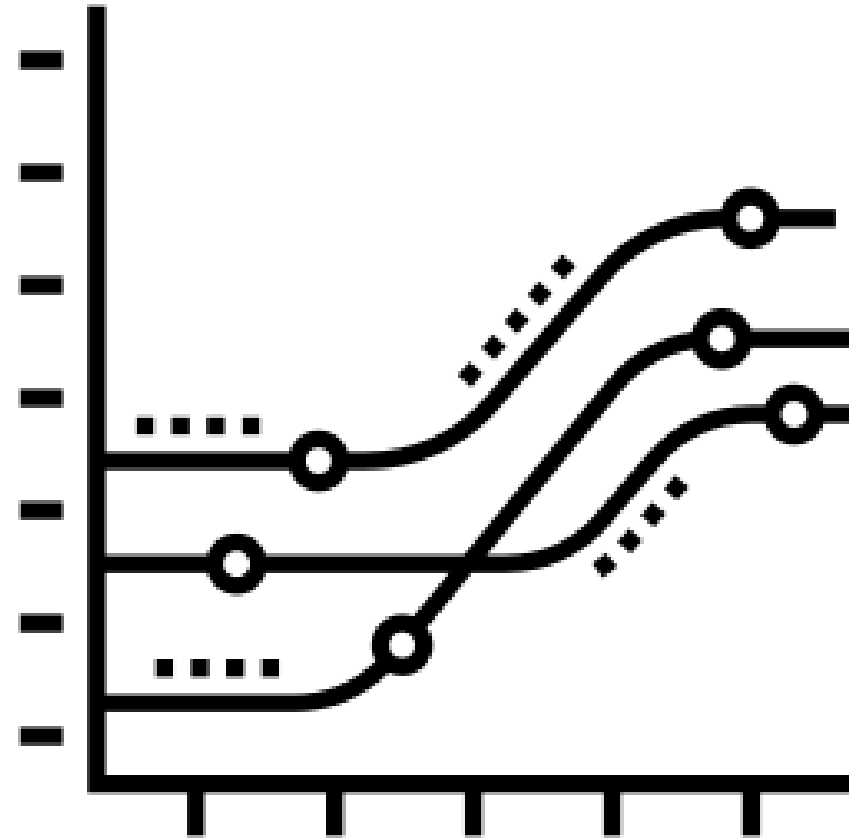


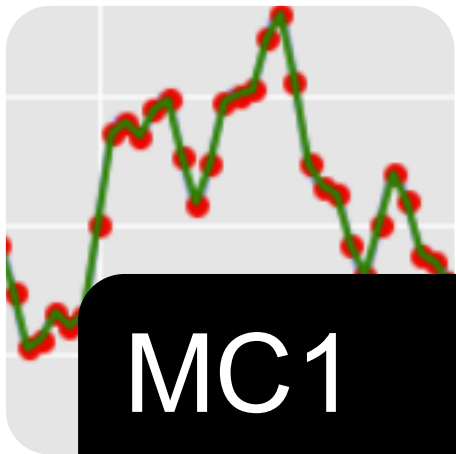
Mini-Challenge Kickoff

gbsv: Foundation in Image and
Signal Processing
FS26

Susanne Suter
3. Mar 2026



Mini-Challenge 1 (MC1): Signal Processing Methods

**MC1**

- 1D
- Signals

**MC2**

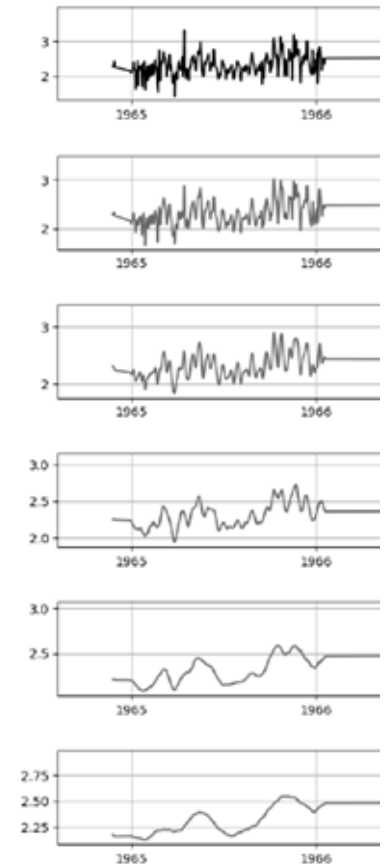
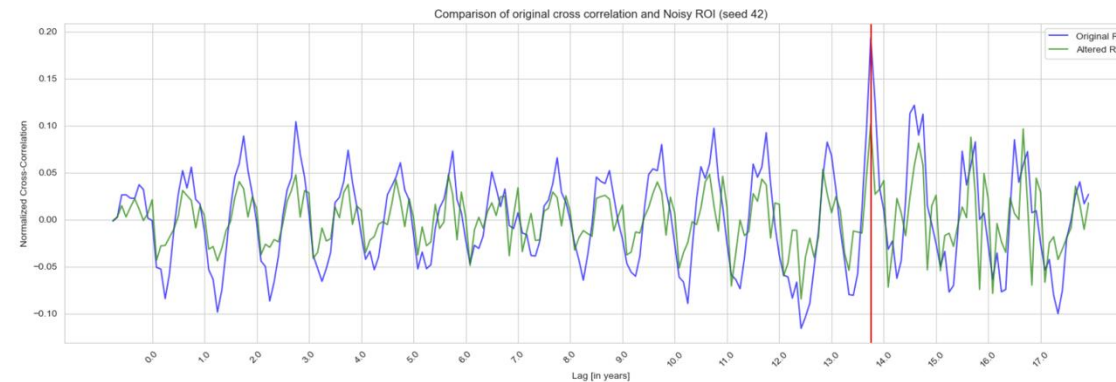
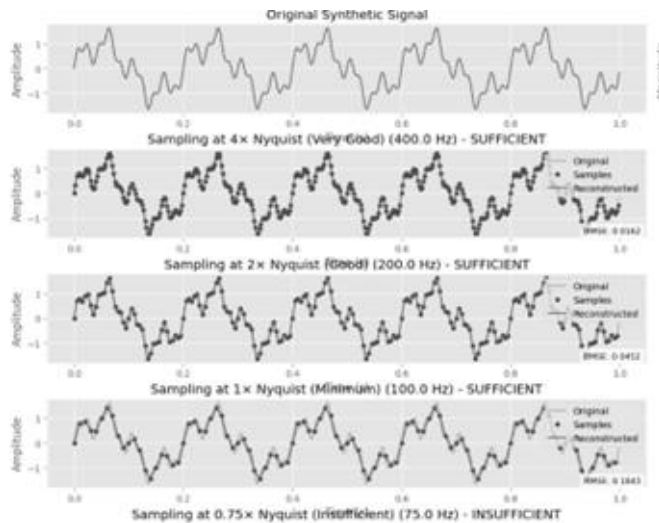
- 2D
- Images

MC1 Topics: Signals

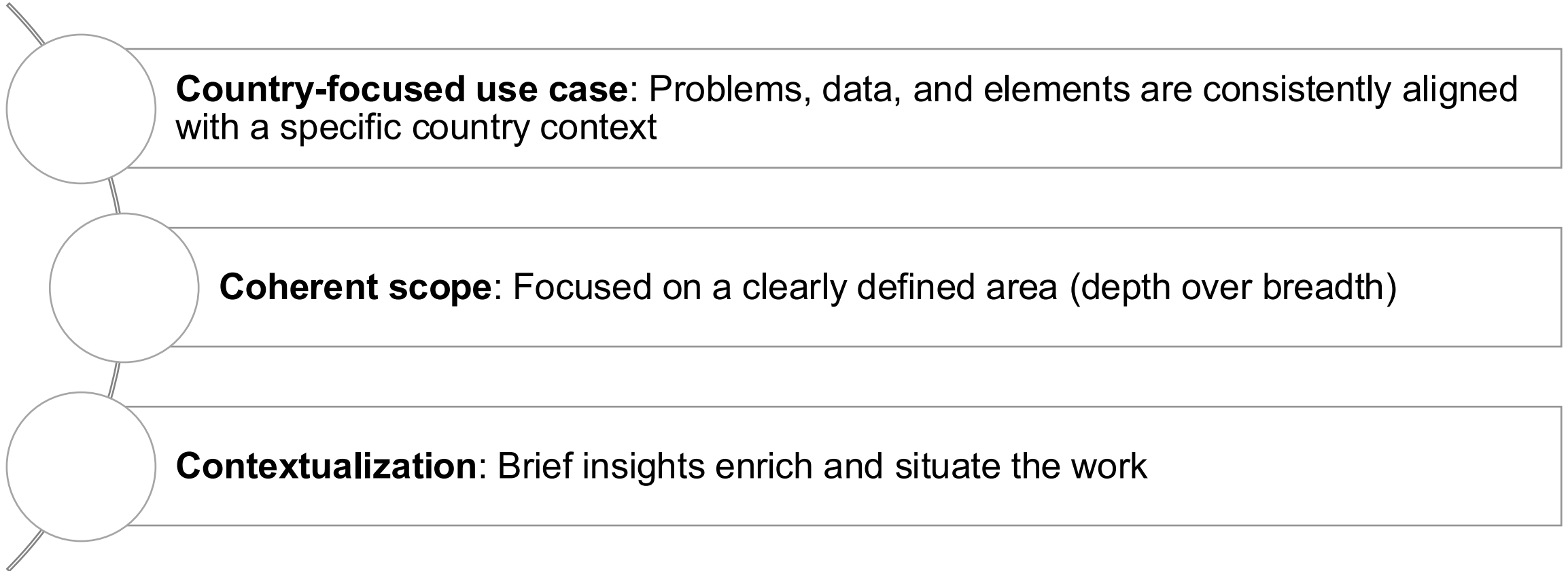
Sampling Theorem

Correlation

Convolution
(Spatial Domain)



Connecting the Dots with a Common Theme of a Country



Inspiration for Swiss Signals



Time Series

- Swiss train timetables (renowned for punctuality)
- Electricity production from Swiss hydro lakes (seasonal peaks in summer meltwater)
- Tourism data (ski resort visitor numbers across seasons)

Signal

- Environmental signals like glacier melt, lake levels, or Alpine temperature curves
- Stock price signals for Swiss companies (Nestlé, Novartis, Ricola exports, etc.)
- Transportation signals - train delays or Zurich airport flight patterns

Audio

- Alphorn sounds (traditional Swiss instrument)
- Cowbells (from Swiss alpine pastures)
- Ambient sounds - mountain streams, cable cars, ski lifts

Use Case and Problem Statement Examples



Time Series

- Use case: Swiss train timetables
- Problem statement: How predictable is Swiss train punctuality over time, and can anomalies (delays) or recurring patterns be detected from timetable data?

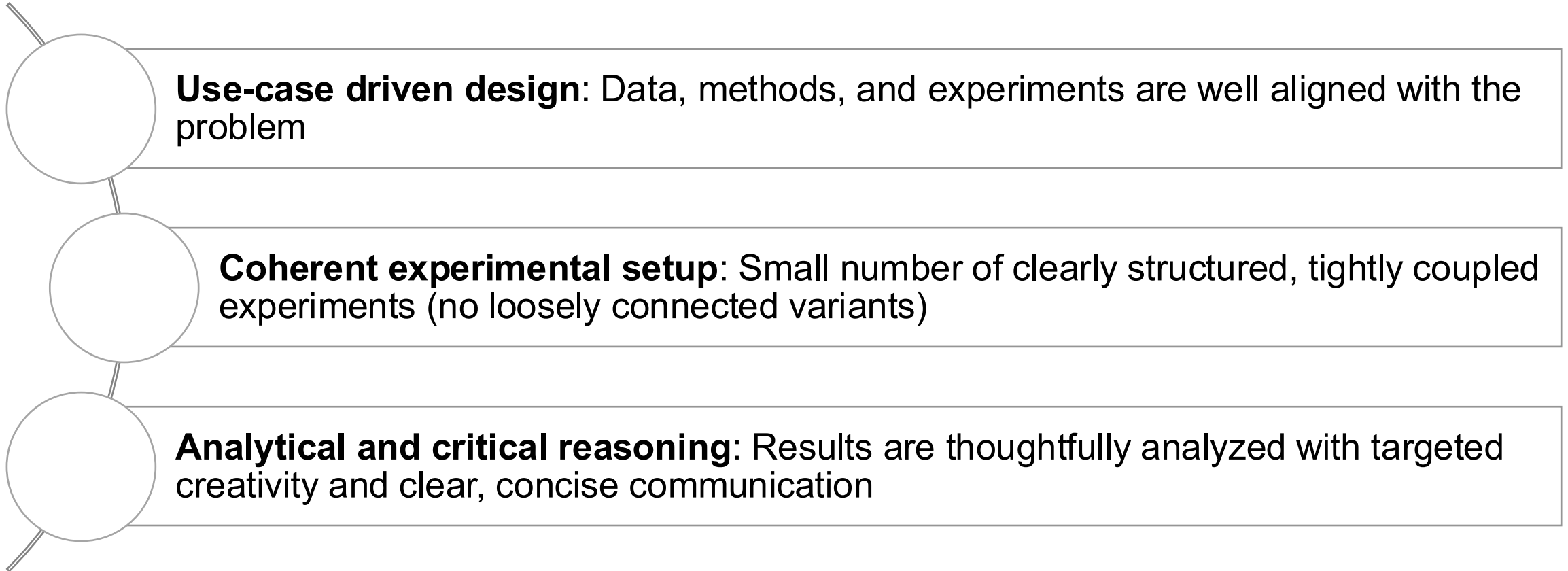
Signal

- Use case: Glacier melt measurements (environmental signals)
- Problem statement: Can signal patterns in glacier melt and lake levels be linked to notable climate events (e.g., heatwaves, heavy snow years)?

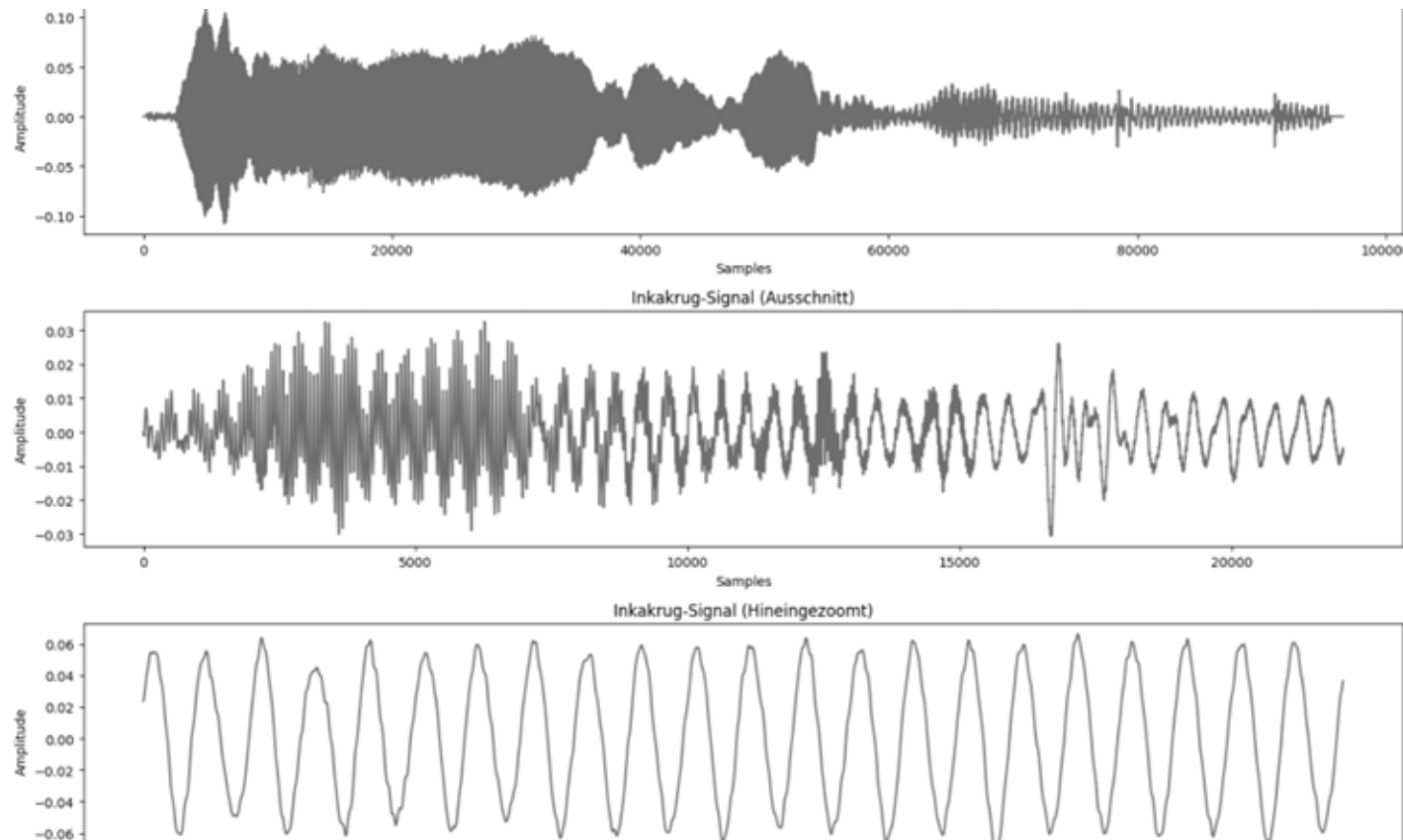
Audio

- Use case: Alphorn recordings
- Problem statement: How can audio signal analysis distinguish between different alphorn tones or playing techniques to preserve traditional Swiss music digitally?

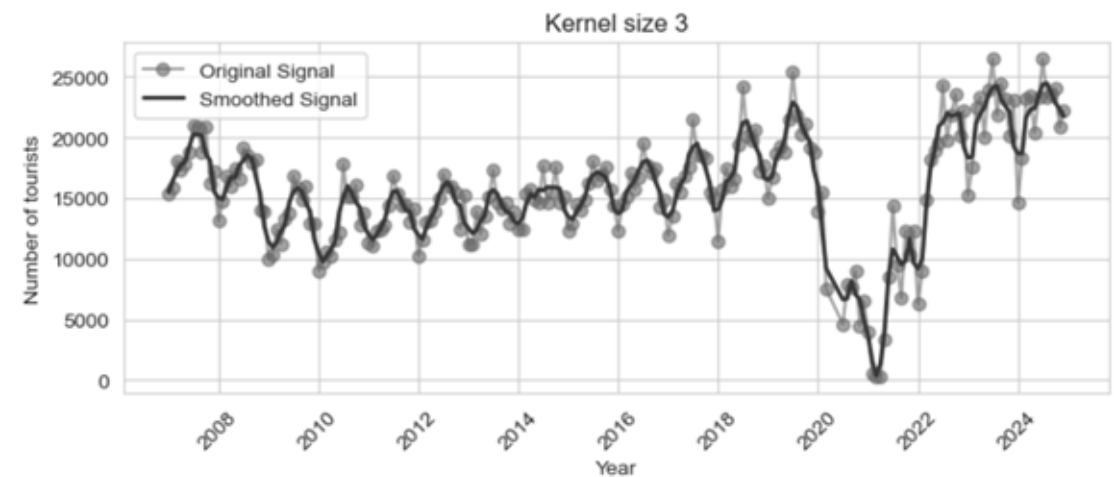
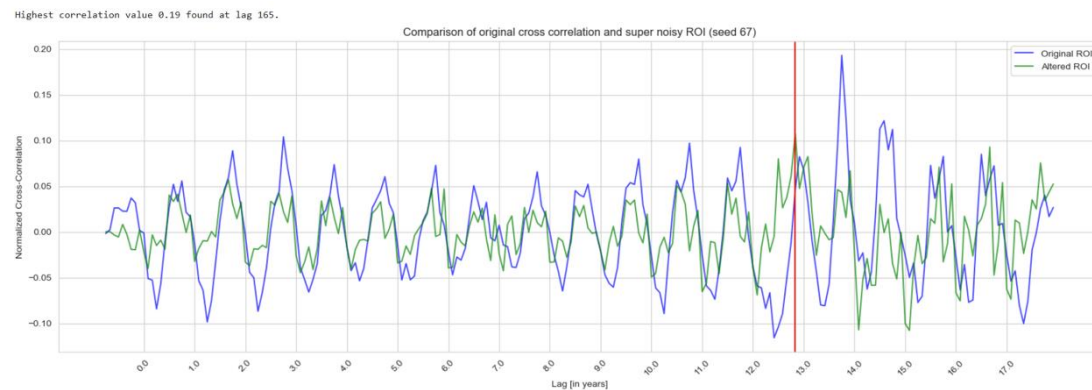
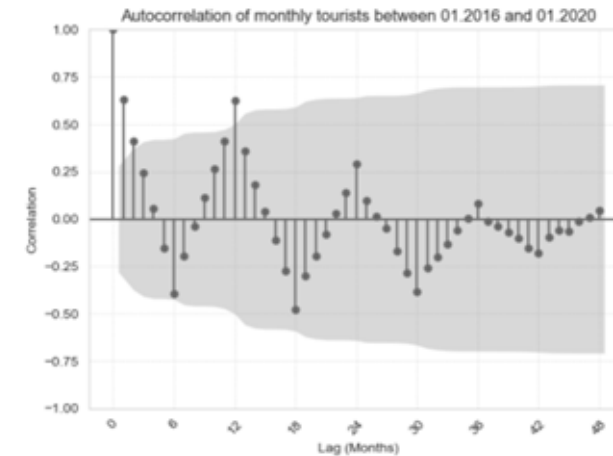
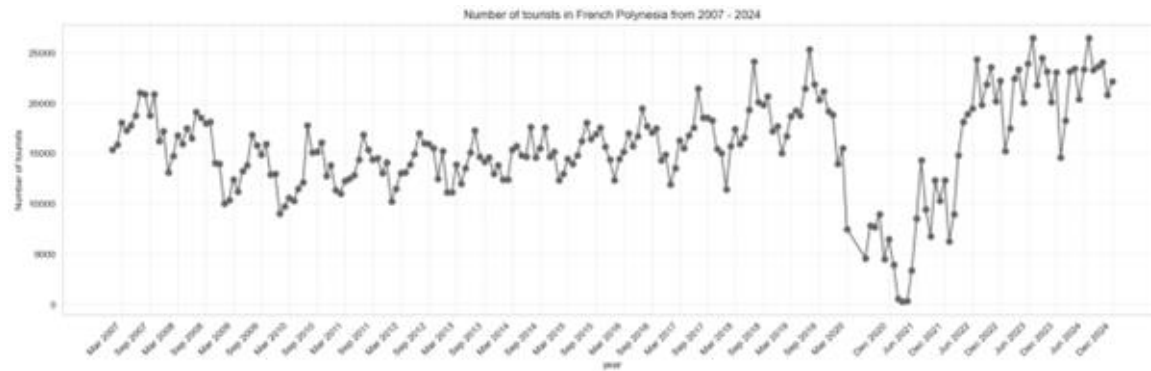
What is important during gbsv MCs?



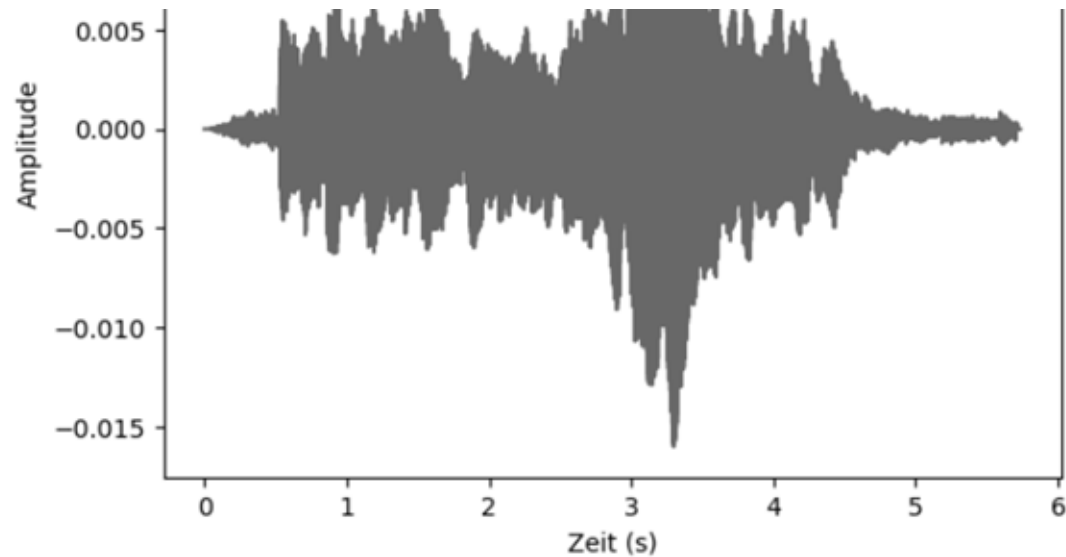
Appropriate Region of Interest Selection



Data May Be Reused for Varying Sub Tasks



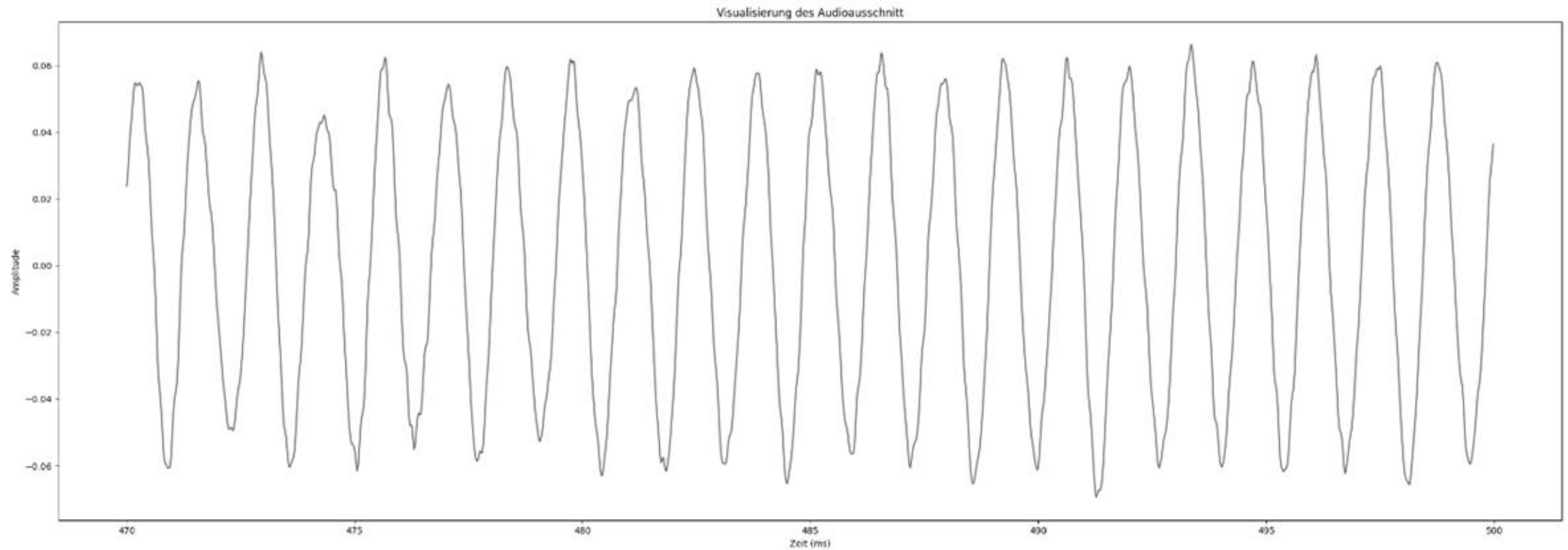
Accompanying KPIs with Figures



```
audio_info(feuerwehr_quiet, sample_rate_f)
```

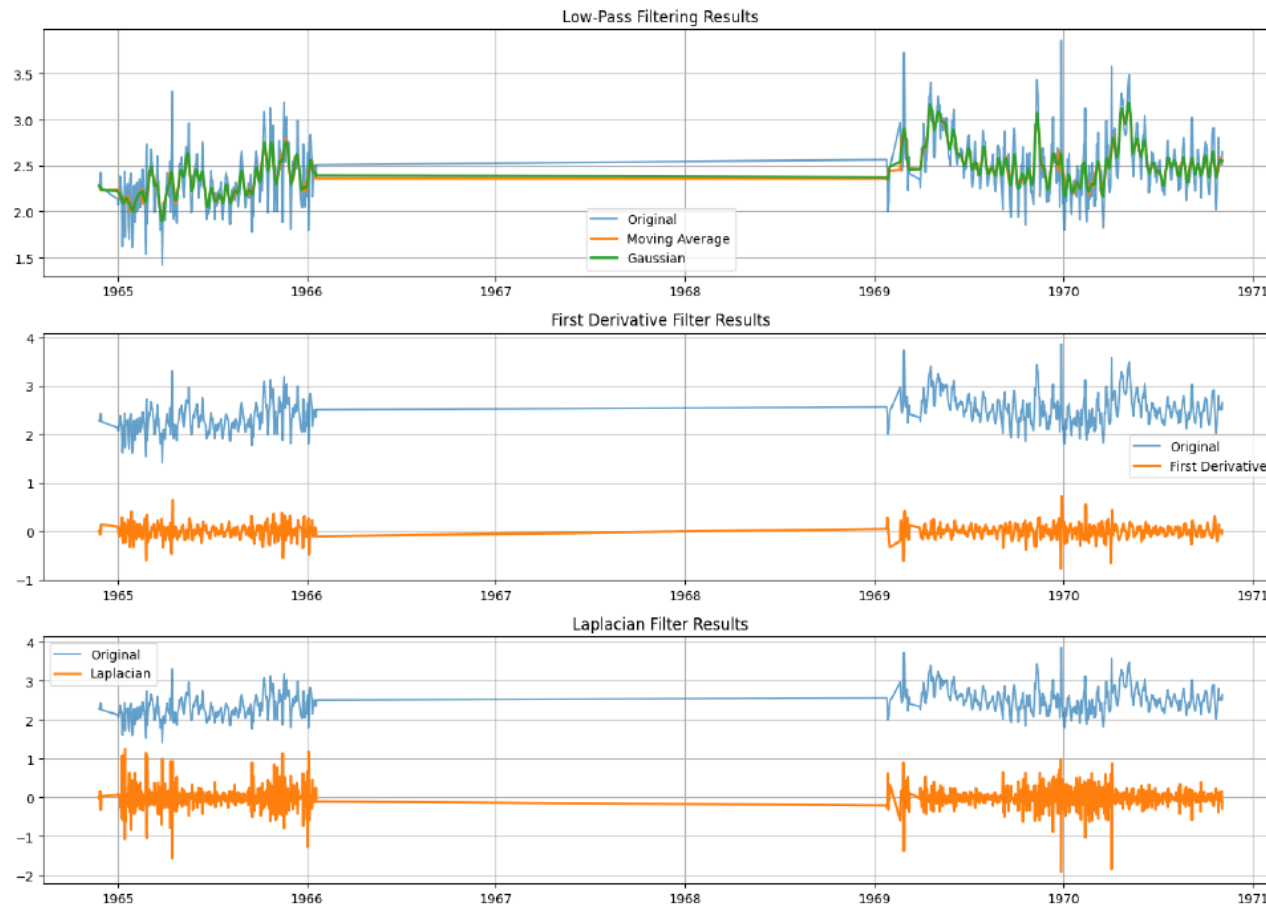
Länge: 126479
Abtaste (Sample Rate): 22050 Hz
Dauer: 5.736009070294784 Sekunden
Amplitudenbereich: -0.0160134956240654 bis 0.014689575880765915
RMS-Rauschen: 0.0023657497949898243

Accompanying KPIs with Figures

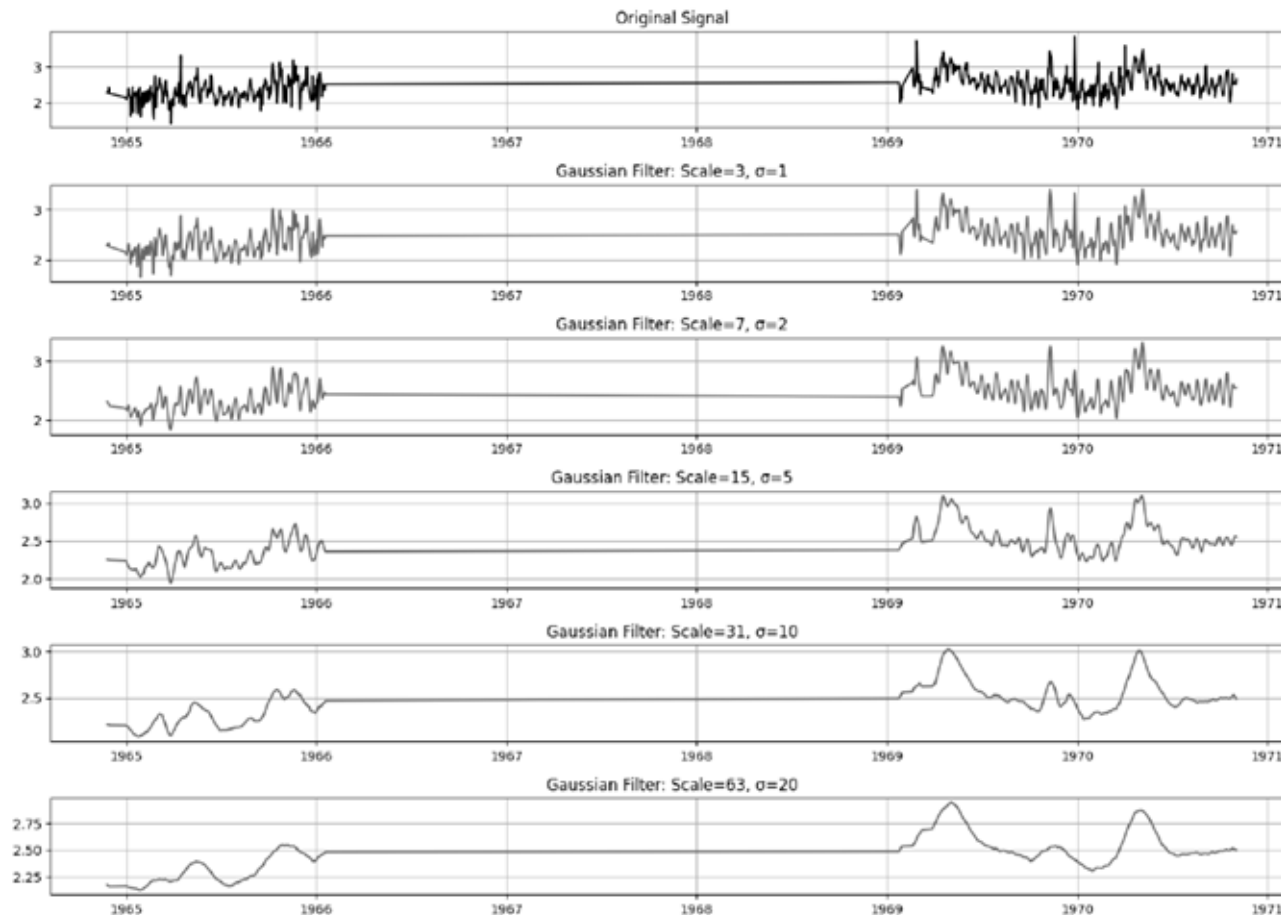


Durchschnittliche Amplitude: 0.06
Durchschnittliche Frequenz: 737 Hz
Durchschnittliche Periodendauer: 1.36 ms

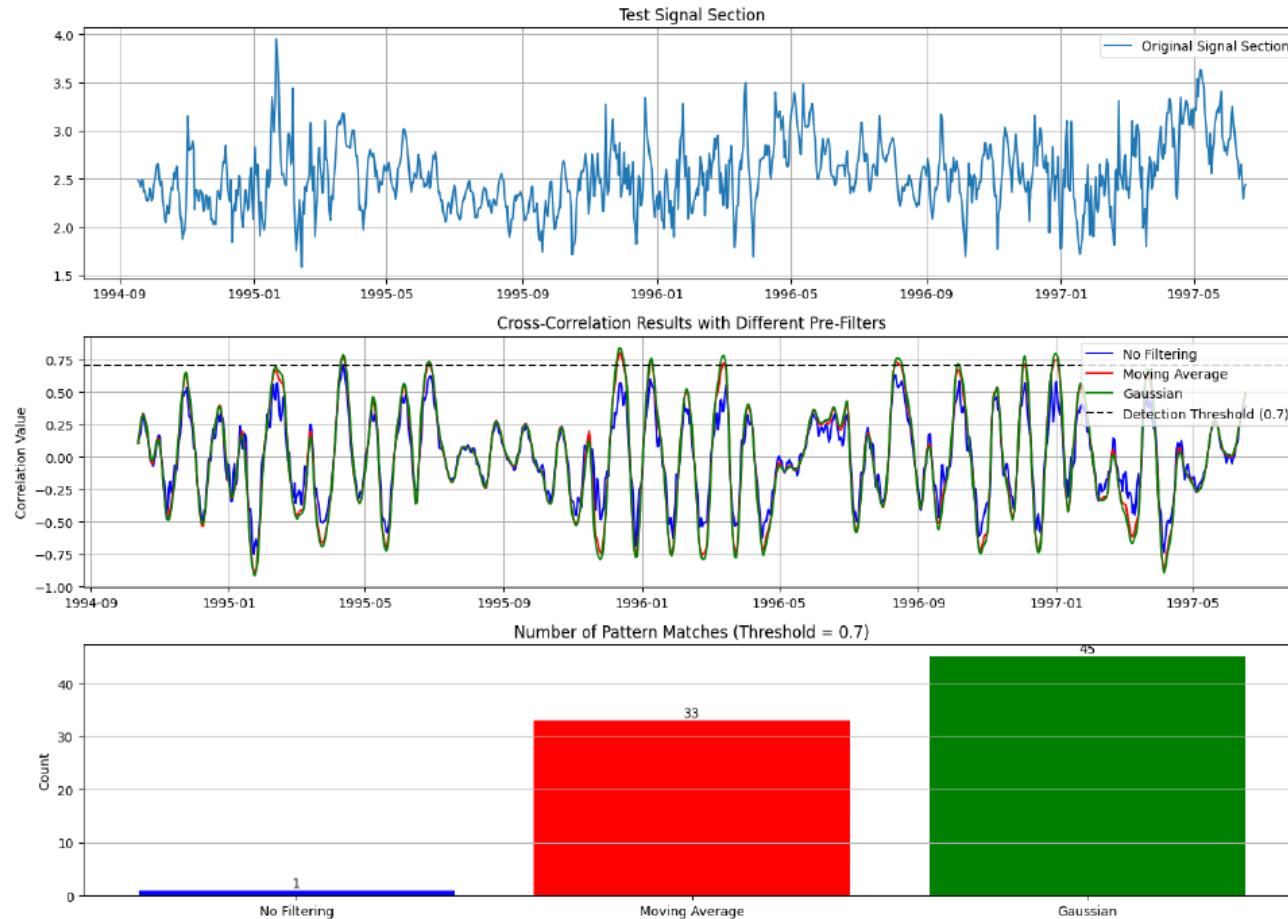
Convolution Examples



Explore Parameter Spaces



Analyze Performance



Mini-Challenge Versions

Version: 15-day-challenges



MC1 15 daily submissions

MC1: individual
assessments

MC2 15 daily submissions

MC2: individual
assessments

kickoff DD signals
MC1 kickoff

MC2 kickoff inputs images



check-in 1

MC1
handin

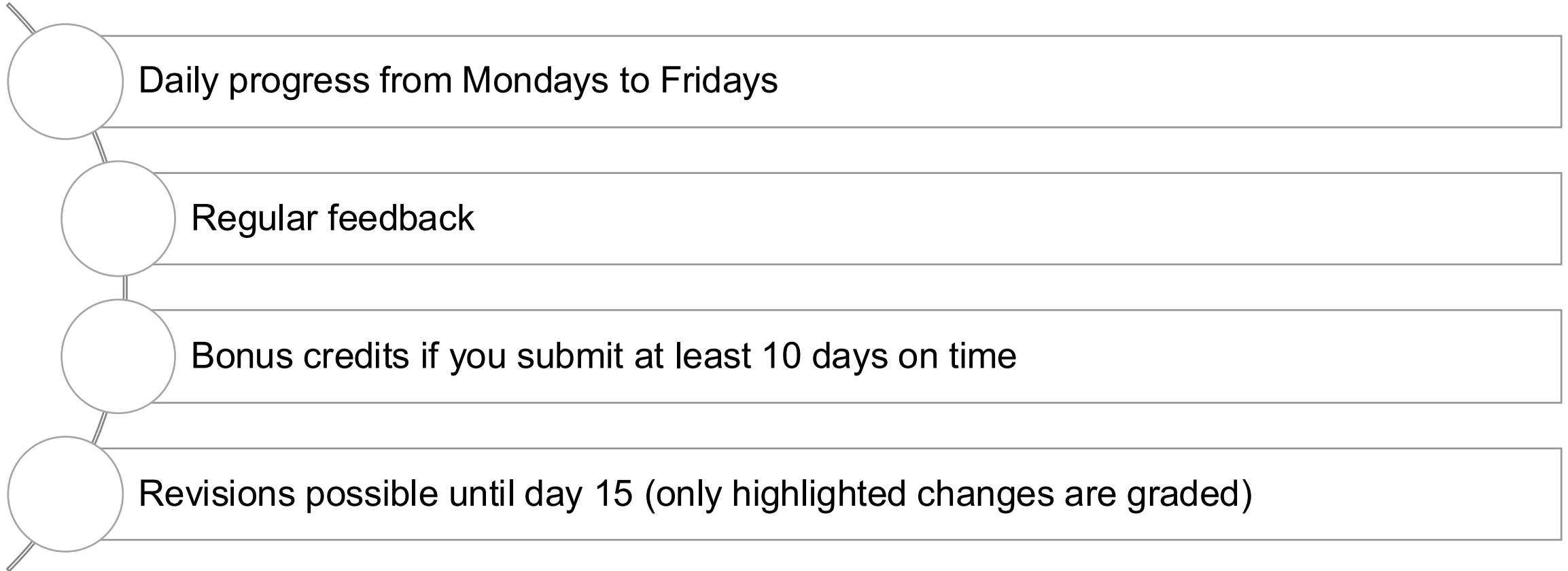
MC2: individual
assessments

check-in 2

MC2: individual
assessments MC2
handin

Version: individual MC rhythm

15-Day-Challenges Per Mini-Challenge

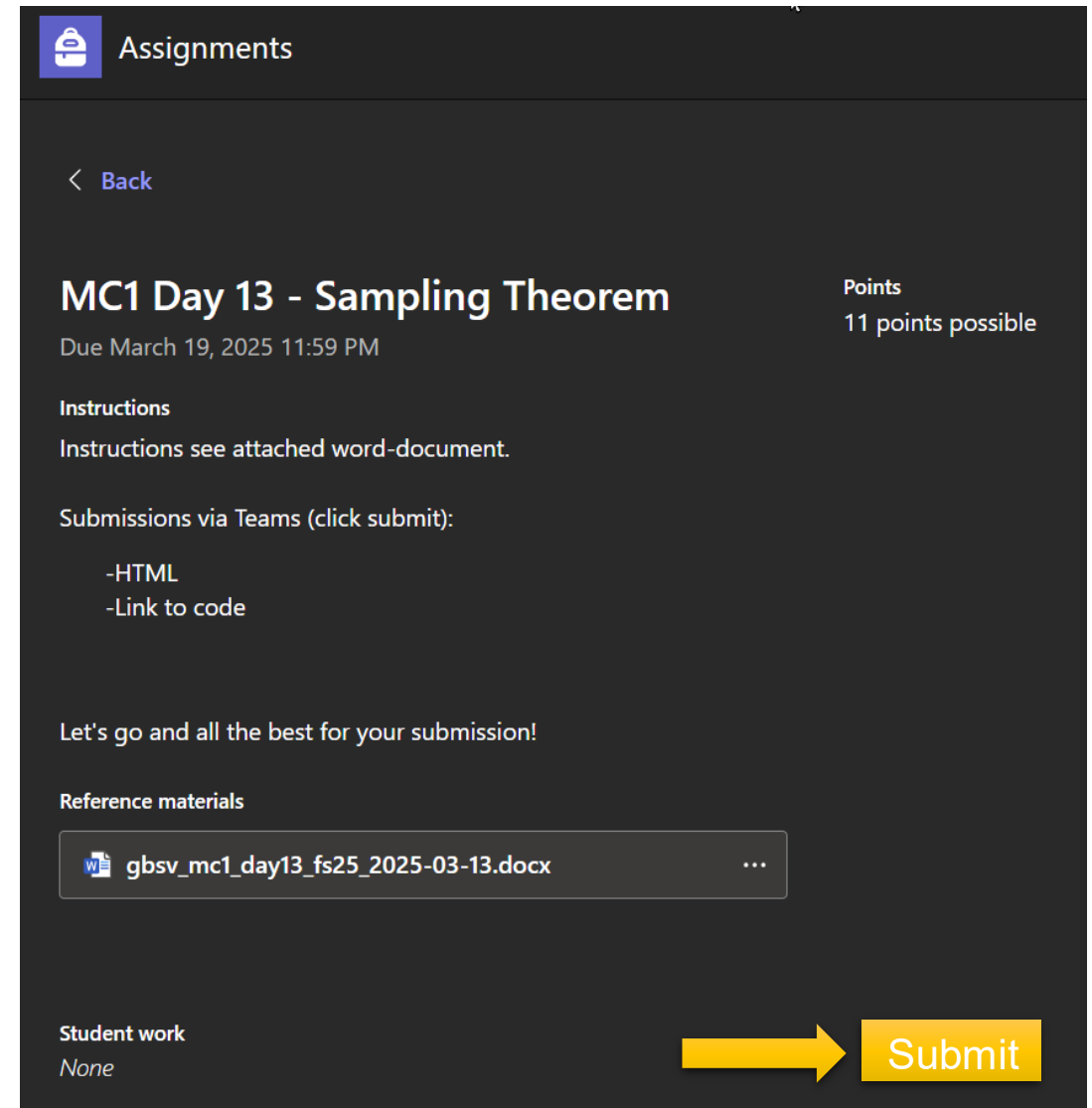
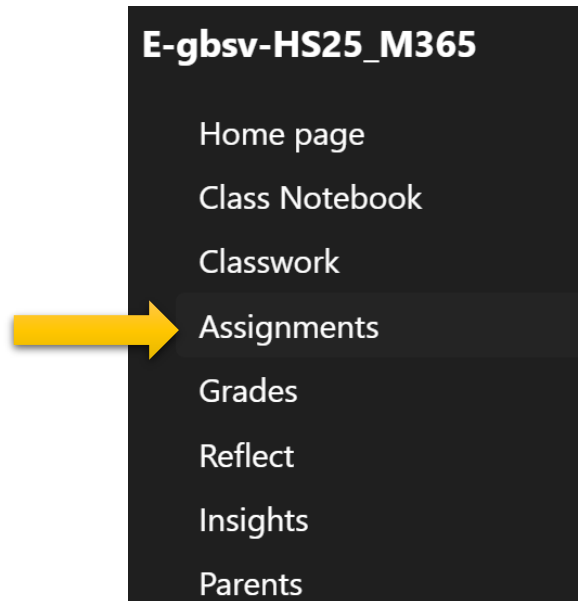


Check-Ins

- **Mandatory for individual MC rhythm**
- 15 min review meetings (see [Video: Warum Check-Ins \(German\)](#))
- One per MC and a third one if things don't go smooth
- During contact hours with subject matter expert → enrollment via [Calendly](#)
- Optional in 15-day-MC versions

Assignments - Submissions

- Submit HTMLs and link to sources in designated Teams assignment: E-gbsv-FS26_M365



Submission Requirements

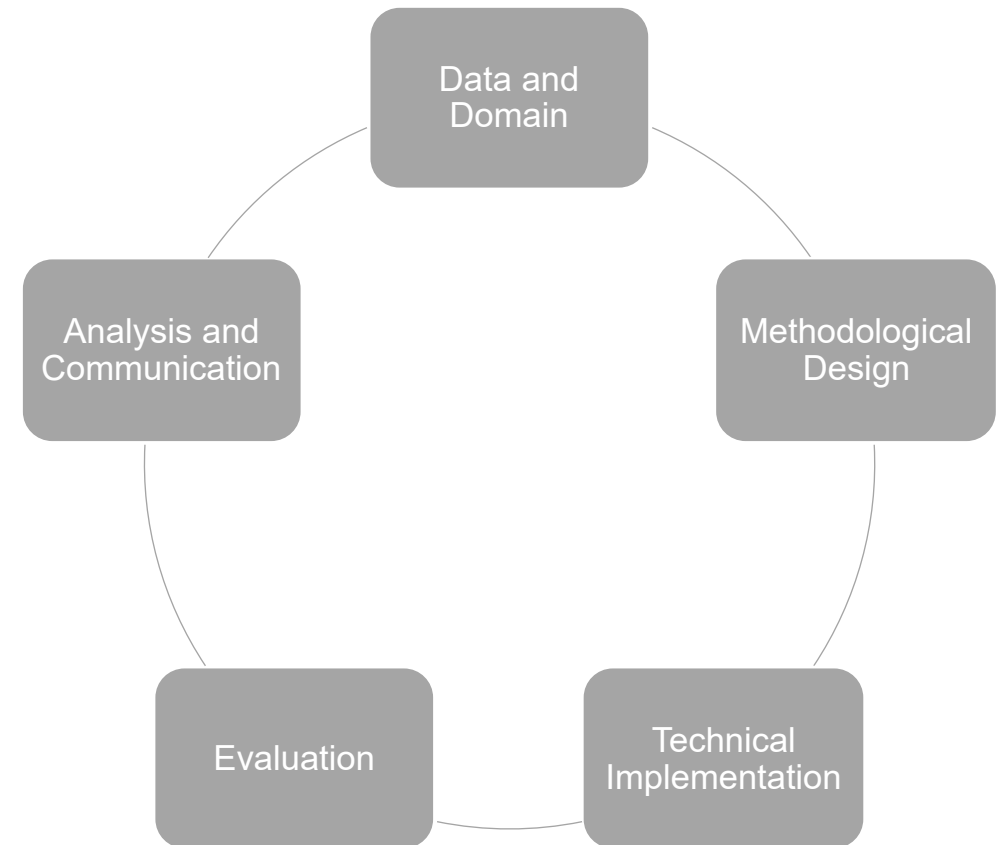
- **Principle:** Individual, personalized work (no generic or shared submissions)
- **Artifacts**
 - Notebook (.ipynb) and HTML export (one notebook per subtask)
 - Accessible link to code and results (incl. signals/images, if applicable)
- **Format:** Language: German or English
- **Acceptance Criteria (mandatory):**
 - No ZIP files
 - No late submissions
 - Check-in completed (individual rhythm)
 - Submissions violating these are rejected

Overall Assessment

- Module assessment: 2 equally weighted parts (signals, images)
- Each part = 50% MC submission + 50% individual assessment
- Final grade = average of Part 1 and Part 2

MC-Grading

- According to 5 competences
- Submission rejected when:
 - Late
 - No check-in per MC



Individual Assessments: Grading

- 6 questions, each rated with 0-2 points → max. 12 points
 - 2: correct, precise, can explain why
 - 1: partially correct / understands basic idea but explanation incomplete
 - 0: incorrect / cannot explain
- **Point-to-grade conversion:** $\text{Grade} = 1 + 5 \cdot \frac{\text{Points}}{12}$, rounded to 0.1; 12 points = grade 6.0, 7 points = grade 3.9
- **Keep answers concise:** aim for approximately 1–2 minutes per question; I may interrupt once I have sufficient evidence and move on to the follow-up question. You can also say you want to skip an question.

15-Day-Challenges Per Mini-Challenge

- Conduct the mini-challenge two 15-days-challenges and profit from:
 - Daily progress from Mondays to Fridays
 - Regular feedback
 - Earn 1 bonus credit per submitted day if you submit **at least 10 days**
 - Bonus credits: up to +0.33 added to final grade (scaled to 15; capped at 6.0 for max grade)
 - Submission deadline: all entries must be submitted by Day 15 (per challenge)

Individual Assessments per MC: Style

- **Goal:** Do you understand what you did, why it works, and what happens when the situation changes?
- **Format:** 15 min/student, oral, closed notes, based on your individual MC submission
- **3 different topics** (e.g., Sampling Theorem, Correlation, Convolution)
 - one main question + one follow-up question per topic
- **Question types:**
 - **Explain:** “You did X here. What does this result mean?”
 - **Justify:** “Why did you choose X rather than Y?”
 - **Transfer:** “Suppose I change X to Y. What would you expect to happen, and why?”

How much time should I spend on daily submissions?

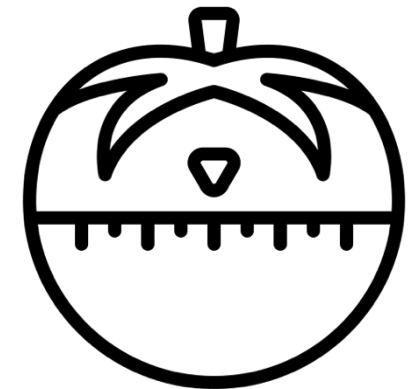
- Certain tasks are intentionally kept open-ended
- Set your own frame for limitations, or ask if the boundaries are unclear
- As a rule of thumb: one day's task should take around 2 hours (+/- 30 minutes)
- Extensive submissions/texts are neither required nor expected



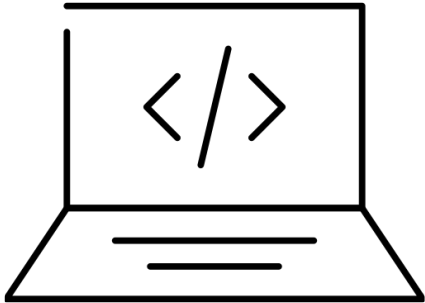
The Pomodoro Technique

- Work in focused intervals: 25 minutes (“1 Pomodoro”)
- Take a short break: 5 minutes after each Pomodoro
- Repeat: 4 pomodoros in a row
- Longer break: 15–30 minutes after 4 pomodoros
- Goal: Boost focus, reduce fatigue, and keep productivity high

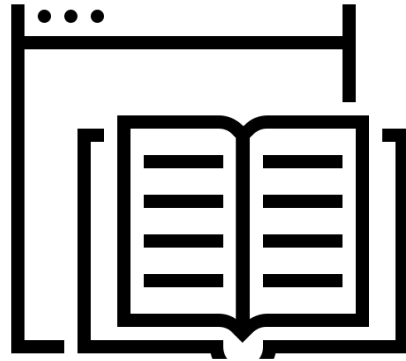
- MC:
 - Stop after 3-5 pomodoros - don't overwork in one go
 - Keep your submission time-boxed - focus on essentials
 - Revise later - e.g., on day 15 after a break



Suggested Resources: your Swiss pocket knife for the MC



[Tutorial notebooks](#)



[Digital Signals Theory Website](#)



[FAQ videos](#) (German)



[Lesespuren](#): escape room style short learning quizzes (German)

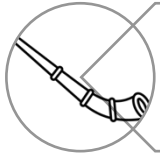
A Recipe for MC Quality, à la Swiss-made



Glide through your assignment like a skier: smooth, structured, and controlled



Match data, methods, and problems like fondue: a perfect fit



Let your creativity echo like an alphorn: focused and distinctive



Sharpen your analysis like a Swiss watch: precise and reliable



Keep it solid like Swiss cheese: clear, structured, and traceable

Acknowledgments

- Signal examples are from previous students taking the MC